

Thinking Sideways, Observed: Interpreting the Lateral-Thinking Search Behavior of LLM Agents

Xiaoxuan Lei

(co-authors TBD)

Draft v0.12 — IAB: Interpreting Agent Behavior, Workshop @ NeurIPS 2026

Abstract

Interactive agents are usually evaluated by endpoint scores, yet a score records where a search ended, not how it moved. We study this gap in Turtle Soup, a lateral-thinking game in which an agent must reconstruct a hidden story through yes/no questions. We ran three Qwen models (4B, 27B, 397B) on 22 human-played puzzles, scoring a forced checkpoint answer every round and embedding each round’s question to form an association trajectory. We found that: (1) gains stop after round ten at every model tier (397B 0.124 $\rightarrow$ 0.200, then flat; 4B flat at 0.05–0.06), and the round budget is not the constraint; the larger tiers are flat across caps 5–30, and the smallest model’s apparent decline under longer caps (0.050 $\rightarrow$ 0.004) is a token-exhaustion harness artifact absent from non-exhausted games; (2) two behavioral patterns emerge, one from trajectory geometry and one from score dynamics and commitment: the small model *circles* (late-game stride 0.030 vs 0.082, robust to excluding malformed rounds; 2/66 games committed), whereas larger models *abandon*, reaching solution-grade checkpoint stories mid-game yet committing a final story in at most 11 of 66 games; (3) within puzzles, mixed-effects models detect no predictive power for stride or drift (coefficients bounded near zero, all likelihood-ratio $p > 0.09$, under both anchors), so the geometry is an interpretive instrument rather than a score predictor; and (4) in a pre-experiment pilot, raising a failing Oracle’s accuracy from 30% to 100% left the smallest Questioner’s score unchanged; the environment, not the agent, had been the bottleneck, so agent claims must begin with an information audit of the environment. We release the harness, a provenance-enforced puzzle set with an under-determination difficulty measure, and a two-part score whose halves correlate only weakly per game ($\rho \leq 0.35$).

1 Introduction

Here is a Turtle Soup puzzle. *A man’s body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.* That is all the solver sees. The hidden story: a hot-air balloon was losing altitude; the passengers threw out their luggage, still could not stay up, and drew lots with matchsticks; the man who drew the short one was thrown out. The solver recovers this story by asking yes/no questions, one per round.

Outcome metrics can say whether an agent got there. On our verified set that story is flat: accuracy rises only in the first ten of thirty rounds, only for the larger tiers, and the round budget is not the constraint (§5). What outcome metrics cannot say is how the agent searched. Two agents scoring 0.0 on the same puzzle may be failing in opposite ways: one never leaves its first hypothesis, while the other finds the answer and lets it go. Telling them apart requires instruments that read the process; providing them is this paper’s contribution.

Why Turtle Soup. The behavior *is* text, so no probing or activation access is needed; puzzles are deliberately under-determined, so each associative jump’s direction is diagnostic; and games are short, language-only, and cheap at scale.

Central hypothesis. An agent’s per-round questions, embedded in a semantic space, form a trajectory whose geometry explains success and failure. **H1 (stalling):** an agent stuck in a hypothesis basin shows its stride (per-round semantic step size) shrinking toward zero, of which lexical novelty heuristics catch only the verbatim extreme. **H2 (drifting):** an agent exploring where no human path goes shows normal stride but growing anchor distance. **H3 (predictivity):** trajectory features predict final solve quality beyond outcome-adjacent covariates. The results confirm H1 (as circling), reject H2’s drift, reveal a second, unhypothesized pattern (abandoning), and do not support H3 (§5).

Contributions. (1) A dissection of the flat curve: the round budget is not the constraint, the smallest model’s apparent budget harm is an environment artifact, and model tier raises mid-game peaks without producing further gains after round ten. (2) Two behavioral patterns, *circling* (from trajectory geometry) and *abandoning* (from score dynamics and commitment behavior), with an honest account of what the geometry does and does not predict. (3) An open-source harness, a **22-puzzle set restricted to puzzles humans have played and solved**, and a composite score decoupling

clue recall from judged causal logic. (4) A puzzle-level difficulty measure, **under-determination**, independent of rounds used, so it can group an accuracy-versus-rounds analysis without circularity.

2 Related Work

Existing turtle-soup benchmarks, TurtleBench [1], SPLAT [2] (whose interactive player–judge protocol we adopt), and TurtleSoup-Bench [3], score verification accuracy, solve efficiency, or faithfulness to the intended story. TurtleSoup-Bench probes what agents ask next but grades it against the intended story; none of the three measures the search’s geometry, which is our delta. Circling and abandoning echo folk-known multi-turn failure modes (repetitive questioning; failing to commit or retain a correct answer); our contribution is to measure them, not to claim their novelty. MUTATE [4] shows that frontier agents find far fewer mechanism-distinct paths than humans, and its diverge-then-narrow agent is a natural intervention to test against our metrics. The instrument itself adapts forward flow [7] and the Divergent Association Task [8] from human psychometrics. The Small World of Words norms [9] provide the human anchor that ours is not yet (§3). Judge-bias catalogues [5, 6] shape the scoring: the model judge is confined to the one subscore that objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the 汤面 (the “surface”, the puzzle’s visible scenario) and asks one yes/no question per round. An **Oracle** holds the 汤底 (the hidden story) and answers 是/不是/与此无关 (yes, no, or irrelevant). The Questioner may commit a final story at any round, or is forced to at the budget. The open-source harness has pluggable model providers and logs every round, with per-game seeds.

3.2 Puzzles

A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set therefore takes only puzzles with a public record of human play: 22 items from a Chinese Turtle Soup community site, each keeping its source URL. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102), with 97 annotated key clues. Provenance is enforced in code, with a verified family, a quarantined generated family, and tests that fail if the two mix.

Each puzzle carries an **under-determination** score: how far the surface alone leaves a solver from the solution, measured by taking the closest of twelve cold guesses written from the surface with no Oracle feedback. It ranges 0.173–0.577 (median 0.347) over the set and is *not* explained by surface length ($r = -0.22$, n.s.). It therefore enters the mixed-effects analysis in §5 as a puzzle-level covariate.

3.3 Outcome scoring: composite, not surface

Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**; Appendix B gives the detail. String matching alone cannot tell a wrong answer from a right one worded differently: we watched a frontier model reconstruct a hidden causal chain completely and score 0.00. Conversely, the clue half blocks “far = creative” gaming, because distance without validity scores zero.

3.4 Association-trajectory instrumentation

Each round, we extract the question’s keywords (jieba), embed them (bge-small-zh-v1.5), and average them to a unit round vector q_t ; distances are cosine, and anchors embed the puzzle’s surface, solution, and clue terms (specification in Appendix B). Two signals follow: the **stride** $s_t = d(q_t, q_{t-1})$ and the **anchor distance** $h_t = d(q_t, \mathcal{A})$. The hypothesized signatures are stalling ($\bar{s} \rightarrow 0$ with h flat) and drifting (normal $\bar{s}$ with h_t rising). The lexical `question_novelty` metric is kept as the ablation baseline that H2 predicts will miss synonym drift.

On the anchor, stated plainly. Ours is built from the puzzle’s surface, solution, and clues; it is **not** a human association manifold, and we no longer describe it as one. Its terms overlap the clue set that defines the outcome. Rebuilding it from the surface alone also flips the drift slope’s sign on a minority of traces (quantified in §5). We therefore report both anchors and treat a SWOW-based version [9] as the correct future form.

4 Experiment Designs

Designs. E1 (round curve) plays to 30 rounds, forcing and scoring a checkpoint answer after each round. E2 (round budget) imposes hard caps {5, 10, 15, 20, 25, 30} with a forced final answer. E3 (association trajectory) computes $\{s_t\}$ and $\{h_t\}$ for every E1 game and tests H1–H3, with the keyword extractor held fixed across models so no model is scored through its own vocabulary.

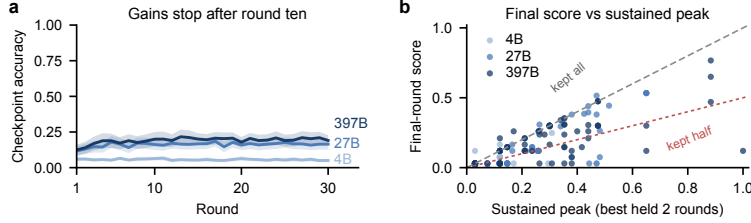


Figure 1: (a) Mean checkpoint accuracy by round (66 games per model; bands, 95% CI), full 0–1 range. (b) Final score against sustained peak (best held two rounds), one point per game; points under the red line ended below half their peak (18–20 of 66 per model).

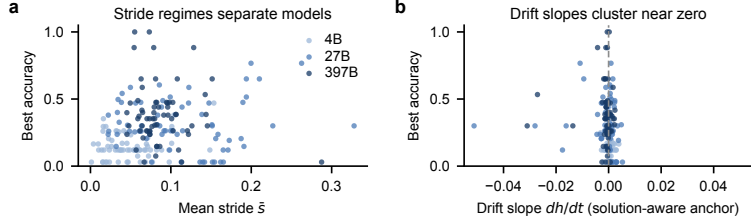


Figure 2: Per-trace geometry against best outcome ($n = 198$). (a) The models separate into stride regimes, but within a model stride does not predict outcome. (b) Drift slopes cluster near zero at every scale: H3’s negative result and H2’s absent signature in one view.

Prerequisite: the Oracle audit. The Oracle is the sole information channel: a weak one collapses answers to “irrelevant”, after which no Questioner can succeed and every score measures the environment. In a pilot, a 4B Oracle (thinking disabled) answered 30% of probes correctly; raising its accuracy to 100% left the 4B Questioner’s score unchanged. Candidates are therefore probed on held-out (question, answer) pairs before any agent claim (protocol and per-candidate numbers in Appendix B), and **must come from a different model family than the Questioners**, or a Questioner’s resemblance to the Oracle cannot be separated from its tier.

5 Results

All numbers come from one grid on the verified set: $22 \text{ puzzles} \times \{\text{Qwen3.5-4B}, \text{Qwen3.6-27B}, \text{Qwen3.5-397B-A17B}; \text{hereafter } 4\text{B}, 27\text{B}, 397\text{B}\} \times 3 \text{ seeds}$, with a DeepSeek-V3.1 Oracle and logic judge (cross-family with the Questioners; audit and settings in Appendix B). The three models differ in generation and architecture (the 397B is a mixture-of-experts with 17B active parameters), so we speak of model tiers, not scale; harness failures were zero, and token exhaustion is a game outcome reported below. Accuracy denotes the composite score rescaled to $[0, 1]$. E1 comprises 198 thirty-round games with a scored checkpoint every round; E2, 1,188 games across caps $\{5, 10, 15, 20, 25, 30\}$.

5.1 More interaction stops helping after ten rounds (E1)

Checkpoint accuracy rises only in the first ~ 10 rounds, and only for the larger tiers: 397B climbs from 0.124 (round 1) to 0.200 (round 10), then stays between 0.18 and 0.21; 27B rises 0.117 $\rightarrow$ 0.162 and plateaus near 0.16; 4B is flat at 0.05–0.06 throughout (Figure 1a). Twenty further rounds of evidence buy no accuracy at any tier.

Commitment separates the models more than accuracy does. 27B volunteers a final story in 11/66 games (mean round 11.1), 397B in 9/66 (14.2), and 4B in 2/66 (23.5); the smallest neither improves nor concludes.

Peaks, finals, and an honest null. Model tier raises what is found mid-game: the *sustained* peak (best score held over two consecutive checkpoints, robust to single-round rewording swings) averages $0.084 / 0.240 / 0.296$ across tiers, while finals sit lower (Figure 1b). Retention, the mean per-game final-to-sustained-peak ratio over games with a nonzero peak (final = last played checkpoint; excluding early-committing games changes it by ≤ 0.03), is 0.69 – 0.76 at every tier (bootstrap 95% CIs $\pm \approx 0.08$). We do not read that gap as evidence of systematic loss: under a within-game permutation null (knowledge constant, only checkpoint-scoring jitter), expected retention is 0.79 – 0.84 and the observed values are not significantly below it ($p \geq 0.09$). The quantitative case for “finding and losing” is therefore inconclusive at this sample size; the evidence for abandonment is the commitment record and the transcripts (§5.3, Appendix B).

5.2 Budget pressure and a harness artifact (E2)

End accuracy is insensitive to the round cap for the larger tiers (397B 0.19–0.21 and 27B 0.14–0.18 across all six caps). The smallest model’s raw curve declines monotonically (0.050 at cap 5 to 0.004 at cap 30), but this is an environment effect, not a behavioral one: games ending by token-budget exhaustion rise from 0/66 at cap 5 to 63/66 at cap 30 (175 of 396 overall, versus 2/396 for 27B and 0/396 for 397B), and conditioned on non-exhausted games 4B’s end accuracy does not decline (0.05–0.09 across caps). By our own audit principle, the raw decline measures the harness. It also resolves an internal check: 4B’s E1 round-30 mean (0.05) and E2 cap-30 final (0.004) disagree exactly because of exhaustion; on non-exhausted games the protocols agree (0.05 vs 0.09), as they do at the larger tiers.

The two score halves dissociate. Every model earns proportionally far more of the judged causal-logic half than the clue-recall half: 397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, and 4B 2.9 against 0.2, and the halves correlate only weakly per game (Spearman $\rho = 0.30 / 0.35 / 0.06$ for 397B / 27B / 4B). Models reach the right causal shape well before, and often without, landing the annotated clue content; the logic half is not redundant with string-level recall.

5.3 Two patterns: circling and abandoning (E3)

Mean stride separates the tiers: 0.049 (4B) against 0.088 (397B) and 0.114 (27B), with 4B’s late-game stride at 0.030, under half of 27B’s 0.082. That is H1’s signature, which we name *circling*. It is not a malformed-question artifact: 17% of 4B’s logged questions are truncated or non-interrogative ($\sim 1\%$ at the larger tiers), yet excluding them moves 4B’s mean stride only from 0.049 to 0.056, still under half of 27B’s. Its sustained peak (0.084) shows it rarely finds a solution-grade story at all. The larger models fail in the opposite way, by *abandoning*: sustained peaks reach ≥ 0.5 in 4–6 of 66 games each, yet a final story is volunteered in at most 11 of 66 (§5.1). Below-half-peak endings alone do not separate the tiers (18–20 of 66 everywhere; the jitter null explains much of that); the separators are whether a solution-grade peak existed and whether it was committed. H2’s drifting signature, by contrast, barely appears: drift slopes cluster near zero at every tier (Figure 2, right); the second real pattern, abandonment, is detected in score dynamics and commitment rather than geometry.

The patterns in language space. Circling is only half-visible to lexical novelty: 4B re-asks earlier questions verbatim in late rounds (caught lexically), but fresh-worded re-asks of one hypothesis read as new, and only the geometry detects them. Abandoning is invisible to lexical measures altogether; Appendix B walks one game whose round-10 checkpoint story *is* the solution, later abandoned.

Pooled across models, stride correlates with best checkpoint score (Spearman $\rho = 0.23$, $n = 198$), a between-model effect; within each model no correlation is detected ($|\rho| \leq 0.16$ for every model and feature; Figure 2). The mixed-effects test makes that precise (best checkpoint accuracy; Gaussian LMM, puzzle random intercept, tier fixed, ML). Stride adds nothing over the covariates ($\beta = -0.010$ per SD, 95% CI $[-0.035, 0.014]$, LRT $p = 0.42$), nor does drift ($\beta = -0.011$, $[-0.035, 0.014]$, $p = 0.39$) or stride $\times$ tier ($p = 0.20$). Under-determination, the puzzle-level covariate, points the expected way (coefficient -0.031) but is not significant ($p = 0.22$) and removes only $\sim 8\%$ of the puzzle variance, so the puzzle effect is not merely measured difficulty. **H3 is not supported within-puzzle**; effects are bounded small but not excluded, and trajectory geometry is an interpretive instrument, one that says *how* an agent fails, not a score predictor.

Both anchors. Drift slopes from the solution-aware and surface-only anchors agree in rank ($\rho = 0.80$) yet flip sign on 38/198 traces. The conclusion is unchanged though weaker under the surface-only anchor (drift $\beta = -0.020$ $[-0.044, 0.003]$, $p = 0.097$): the anchor changes traces, not the finding.

6 Limitations and Conclusion

Limitations. Truncated chain-of-thought under a tight token budget masquerades as an agent that cannot form a question (§5.2, §5.3); decoding configuration is part of the environment. The anchor is not yet human-derived (§3), so H2’s framing is descriptive until SWOW norms or human traces are in place. Keyword extraction and encoders are themselves models, and feedback incorporation, the most direct separator of circling from abandoning, is unmeasured. One DeepSeek model serves as both Oracle and logic judge (a cross-family judge check is pending), and the tiers confound generation and architecture, so tier effects are model comparisons, not scaling laws. The source site is public and memorisation probes are pending (the easiest-rated puzzles are the most circulated; nothing rests on the n.s. covariate). Puzzles involve death and dark themes; content-flagged.

Conclusion. An endpoint score projects a whole search onto one number; circling and abandoning need opposite interventions, and the trajectory tells them apart.

References

- [1] Q. Yu *et al.* TurtleBench: Evaluating Top Language Models via Real-World Yes/No Puzzles. arXiv:2410.05262.
- [2] Q. Chen, B. Zhang, G. Wang, Q. Wu. Weak-eval-Strong: Evaluating and Eliciting Lateral Thinking of LLMs with Situation Puzzles. arXiv:2410.06733.
- [3] M. Zhou *et al.* What to Ask Next? Probing the Imaginative Reasoning of LLMs with TurtleSoup Puzzles. arXiv:2508.10358.
- [4] J. Park, I. Baek, J. Park, H. Lee. Beyond One Path: Evaluating and Enhancing Divergent Thinking in Interactive LLM Agents. arXiv:2605.28465.
- [5] J. Ye *et al.* Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge. ICLR 2025. arXiv:2410.02736.
- [6] L. Shi *et al.* Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. IJCNLP-AAACL 2025.
- [7] K. Gray *et al.* “Forward Flow”: A New Measure to Quantify Free Thought and Predict Creativity. *American Psychologist* 74(5), 2019.
- [8] J. A. Olson *et al.* Naming Unrelated Words Predicts Creativity. *PNAS* 118(25), 2021.
- [9] S. De Deyne *et al.* The “Small World of Words” English Word Association Norms. *Behavior Research Methods* 51(3), 2019.

A The puzzle set

Where the puzzles come from. A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup community: 22 puzzles people have played and solved, each keeping its source record. Every item was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

How hard are they? Each puzzle carries an **under-determination** score: give a model the surface alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to the real solution. The index is one minus that best guess, and is large when the surface leaves a solver far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

The median is 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577: single-peaked with a thin hard tail. Easiest are the 海龟汤 story itself (0.17) and the desert matchstick (0.19). Hardest is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the same* (0.58).

Two properties make this usable as a covariate. It describes the puzzle, not any agent’s performance on it. It is also independent of rounds used; a difficulty defined as “how many rounds this takes” would make an accuracy-versus-rounds plot confirm itself. Nor is it a proxy for surface length ($r = -0.22$, n.s.), so it carries information the obvious surrogate does not. It is one measure rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not separable here.

A caveat we report either way. The two puzzles scoring easiest are the two most widely circulated ones. Their surfaces may be easy to complete not because the inference is short but because the story is in the training data, in which case the measure reads familiarity as much as difficulty. A memorisation probe (asking for the solution with no surface given) separates the two and belongs in any use of this measure.

B Measuring agent performance

What counts as a correct answer. Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and reproducible. The logic judge rates only whether cause → mechanism → outcome matches and is told explicitly to ignore wording. Because single ratings are unstable on borderline answers, ratings are sampled and averaged (two samples per rating in the reported grid). The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be. Below that floor only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above the floor.

Does the logic half earn its thirty points? On the full grid, yes: each model earns a several-fold larger share of the logic half than of the clue half, and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$ for 397B / 27B / 4B over 396 E2 games each). The 70/30 weighting itself is a design choice; every conclusion in §5 is readable from the two halves separately, though the absolute plateau level (~ 0.2) does depend on the weights. The logic judge is sampled twice per rating at temperature 0.2; the two samples agree exactly in 86% of ratings.

Environment settings and Oracle audit. Every model call uses max_tokens 2048 (thinking disabled, temperature 0.2) with a per-game budget of 50,000 tokens; exhausting the game budget forces a final answer and is recorded as the game outcome token_budget. The Oracle audit probes a candidate with 25 hand-crafted (question, expected-answer) pairs over three puzzles, 21 with a definite yes/no answer and 4 expecting “irrelevant”; the pass bar is 90% on the yes/no items. The deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no and 3/4 exact on “irrelevant”. Candidates measured in the pilot phase: a 4B Oracle with thinking disabled, 30% (rejected; this is the configuration behind abstract finding 4, where raising Oracle accuracy from 30% to 100% left the 4B Questioner’s score unchanged); the same model with thinking enabled, 70% (rejected); a same-family flash model, 90% (excluded for sharing the Questioners’ family); GLM-5.3-flash, 100% (used in earlier tooling).

Trajectory and difficulty pipeline. Keywords are extracted with jieba after stripping Latin characters and digits; each keyword is embedded with bge-small-zh-v1.5 (normalized), and the round vector is the re-normalized mean of its keyword embeddings. All distances are cosine. The solution-aware anchor embeds the puzzle’s surface, solution, and clue terms; the surface-only anchor embeds surface terms alone. The under-determination measure prompts GLM-5.3-flash (cross-family with all Questioners) for twelve cold guesses from the surface and scores closeness as embedding similarity between guess and solution under the same encoder, not with the composite score, so the covariate is not coupled to the outcome apparatus.

Artifacts. Harness, puzzle set with per-item source records, per-game logs (including every checkpoint story), and the analysis scripts behind each reported number are released at github.com/RainyFields/Project_Turtle_Soup. Puzzle surfaces and solutions originate from a public community repertoire and are redistributed with source attribution and a content flag.

Where the accuracy curve comes from. E1 does not score once at the end. Every round, after the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored. Each point on the accuracy curve is therefore a full answer written with the evidence available at that round.

One game, worked. Puzzle refsoup_021 (surface: a stormy night, a run-down temple, two photographs on the wall that seem to stare; by morning the narrator sees only two windows; solution: 墙上根本没有照片，是有人贴着窗户往里看, “there were never photos on the wall; people were pressing against the windows, looking in”), played by Qwen3.5-397B, seed 2. Table 1 shows seven of thirty rounds; Chinese cells are verbatim from the log with our translations, stories abridged to the clause carrying the hypothesis. Rounds 22–30, eight lexically distinct re-asks of the same flooding hypothesis (e.g. round 25, whether the narrator woke floating upside down; round 28, whether rising water reached eye level), are in the released logs.

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	照片里的人吗? <i>Are the ones in the photos people?</i>	是 <i>yes</i>	那两张“照片”……是两个被钉死或封在墙洞里的人 <i>The two “photos” are two people nailed or sealed into wall niches.</i>	0.12 (0+12)
3	主角第二天醒来时，还在那座寺庙里吗? <i>On waking the next day, is he still in the temple?</i>	是 <i>yes</i>	所谓的“两张照片”其实是两扇窗户……误看成了照片 <i>The “two photos” are really two windows, mistaken for photos in the dim light.</i>	0.91 (70+21)
4	醒来看到的“两扇窗户”就是昨晚的“两张照片”吗? <i>Are the “two windows” the same objects as last night’s “photos”?</i>	是 <i>yes</i>	“照片”实际上是两个被挖空的人形窗口……恐惧感源于黑夜和错觉 <i>The “photos” were hollowed-out window openings; the fear came from darkness and illusion.</i>	0.12 (0+12)
10	“两扇窗户”是因为相框玻璃破碎脱落吗? <i>Did frame glass break and fall, leaving open holes?</i>	不是 <i>no</i>	那根本不是照片，而是……贴在窗玻璃外侧的人脸 <i>They were never photos, but faces pressed against the outside of the windows.</i>	1.00 (70+30)
11	是因为照片被取走、露出了后面的窗户吗? <i>Were the photos removed, revealing windows behind?</i>	不是 <i>no</i>	昨晚“盯着他”的人像，其实是窗外的人 <i>The staring figures were people outside the windows.</i>	1.00 (70+30)
19	人像是躲在照片后面的真人吗? <i>Were real people hiding behind the photos?</i>	不是 <i>no</i>	窗户洞口被画着人像的布帘封住……雷雨导致脱落 <i>Window openings sealed by painted cloth covers; the storm tore them off.</i>	0.12 (0+12)
30	是因为积水使主角身体位置升高吗? <i>Did rising water lift the narrator higher?</i>	不是 <i>no</i>	两个被纸糊住的窗户洞口……雷雨导致纸张脱落 <i>Papered-over window openings; the storm stripped the paper away.</i>	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Italics are our translations.

Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowledge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from earlier answers on a round whose own question was answered 不是. Yet the model never volunteers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes. **Stalling in fresh words happens at the largest scale too.** Rounds 22–30 are eight lexically distinct re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve’s round-to-round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded (rounds 3 → 4) swings the clue half 70 → 0. The mean curves of §5 average this out, which is why single-game curves are not the unit of analysis.